# ELEMENTS 5

**Elements** are pure substances that contain only one type of atom. Elements can be metals, non-metals or semi-metals (metalloids).

1 Name two metal elements. _____ _____

2 Name two non-metal elements. _____ _____

Coins are usually made of metals. Silver (Ag), copper (Cu) and gold (Au) have been used to make coins for thousands of years. These metals are also used to make jewellery.

3 Give two reasons why jewellery is made from silver, copper and gold.

_____

4 Give two reasons why coins are made from metal.

_____

Most metals are lustrous (shiny) and they are often silver in colour. Metals are good conductors of heat and electricity, whereas most non-metals are poor conductors.

5 Name one metal that is not silver in colour. _____

Metals are malleable (can be beaten into shape). They are also ductile (can be drawn out into wires). Non-metals are brittle (they snap instead of bend). Metals are usually denser (more matter is compressed into a space) than non-metals. Metals are solids at room temperature, with the exception of mercury (Hg). They usually have high melting points. Metals are found on the left side and centre of the Periodic Table.

6 Where in the Periodic Table do you expect non-metals to be located? _____

_____

7 Give another name for *metalloids*. _____

8 Name two metals that are drawn into wires. _____ _____

9 'Most metals are liquids at room temperature.' True or false? _____

10 'Elements contain only one type of atom.' True or false? _____

11 'Non-metals are usually denser than metals.' True or false? _____

12 'Semi-metals have properties of both metals and non-metals.' True or false? _____

13 Name each of the following elements. State whether each element is a metal or a non-metal.

a Al _____

b B _____

c C _____

d Cl _____

e F _____

f H _____

g N _____

h Ne _____

i S _____

j Ar _____

k Be _____

l Ca _____

m Cu _____

n Fe _____

o He _____

p Na _____

q Ni _____

r Si _____

14 Which metal is the only metal that is a liquid at room temperature? _____


978 1 92561 065 9
© Peter Gribben, Wendy St Germain and Blake Education 2005

Some metals, such as iron (Fe), cobalt (Co) and nickel (Ni), can be attracted by magnets or used to make magnets. Metals make up about 25% by mass of the Earth's crust (includes oceans and atmosphere).

**15** For each statement, choose and write the correct word from the brackets.

- **a** Something that is shiny is said to be (lustrous/dull) _____.
- **b** A pure substance that contains only one type of atom: (compound/element) _____
- **c** Can be beaten into shape: (malleable/solid) _____
- **d** Can attract another metal: (electric/magnetic) _____
- **e** Can be drawn into wires: (ductile/liquid) _____
- **f** Snaps when bent: (conduct/brittle) _____
- **g** Allows energy such as heat to pass through easily: (conducts/melts) _____

Sometimes elements chemically join with each other to form **compounds**. These compounds have different properties to the elements from which they were formed.

Often metals combine with non-metals. The name of the compound formed comes from the elements from which it is made. If a compound only contains two elements, its name ends in '-ide'. In a compound that contains a metal and a non-metal, the metal is first in the compound's name. For example, iron oxide (FeO) contains the metal iron (Fe) and the non-metal oxygen (O); potassium fluoride (KF) contains the metal potassium (K) and the non-metal fluorine (F).

**16** For each compound, give the names and chemical symbols for the elements present.

- **a** copper chloride _____
- **b** iron oxide _____
- **c** sodium chloride _____
- **d** lead iodide _____
- **e** magnesium oxide _____
- **f** barium sulfide _____
- **g** calcium carbide _____
- **h** hydrogen bromide _____

**17** Name one chemical compound not mentioned on this page. _____

**18** Give the names and chemical symbols for the elements present in your answer to Q17.

_____

There can be more than two elements present in a compound. If one of these elements is oxygen, the compound's name usually ends in '-ate'. For example, zinc sulfate ($ZnSO_4$) contains zinc (Zn), sulfur (S) and oxygen (O).

**19** For each compound, give the names and chemical symbols for the elements present.

- **a** calcium sulfate _____
- **b** copper carbonate _____
- **c** silver nitrate _____
- **d** sodium phosphate _____
- **e** nickel nitrate _____
- **f** zinc sulphate _____
- **g** lead carbonate _____

© Peter Gribben, Wendy St Germain and Blake Education 2005
978 1 92561 065 9